

The Degtyarev–Shimada primitivity criterion beyond the published table: sixteen verdicts on an 8 GB laptop, and the structure of the CRT splitting

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Abstract

This note summarises an independent computational and structural study of the primitivity criterion of Degtyarev and Shimada [1] for even-dimensional complex Fermat varieties. Three contributions are reported. **(i) An extended census.** The criterion has been verified, byte-exact and with both halves computed independently, in sixteen cells (n, m) — eleven of them beyond the cases reported in [1, §5], including dimensions $n = 8, 10$ and composite degrees up to $m = 14$; the five cells inside the published table are reproduced exactly by an independent engine. Every verdict is `PRIMITIVE`; each is therefore a further confirmed instance of Conjecture 1.2 of [1]. The closed-form rank of [1, Remark 4.4] is additionally verified byte-exact across a 38-cell sweep, well past the published range. **(ii) Structure inside the criterion’s modular half.** For $p \mid m$ the per-variable ring splits by CRT; we prove that any $\mathbb{Z}$ -torsion of the cycle quotient localises to a single CRT block (by centrality of the idempotents), that the rank each block contributes is exactly a Degtyarev–Shimada rank at reduced dimension, $\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1)$, unconditionally — the integral block closure is shown $\mathbb{Z}$ -free via the cyclotomic identity $x \cdot c_0 = -1$, and the CRT idempotent cut is shown to *equal*, as a linear projection, the coordinate-zeroing cut $X(2s)$ of [1, §4.6] — and we record a boundary of the fast Jordan-form reduction (valid precisely for prime-power degree) together with its constructive use as a memory lever. **(iii) The exact relation to Conjecture 1.2.** By the equivalence stated in [1, §4.6], the freedom of the coupled module $(\bigoplus \bar{R}_J)/\langle \sum 1_J \rangle$ in recursive dimension $2s$ is Conjecture 1.2 in dimension $2s$; with the cut identity above, this applies exactly to the blocks measured here. The conjecture itself is untouched. All computations ran on a single 8 GB consumer laptop, single-threaded, in exact arithmetic; the deepest reduction held 3.45×10^9 nonzero entries at 1.07 bytes per entry. Every number is reproducible from the public repository (engines, raw logs, and self-contained verifiers).

1 The census: sixteen complete verdicts

For each cell both halves of the criterion of [1, §5] were computed independently: $\dim_{\mathbb{C}}$ (cross-checked over several primes $P \equiv 1 \pmod{m}$) and $\dim_{\mathbb{F}_p}$ for *every* prime $p \mid m$, by exact sparse Gaussian elimination over $\mathbb{F}_p$. A verdict is recorded only when all halves agree. The published computational record in [1, §5] covers $(4, m)$ for $3 \leq m \leq 12$, together with $(6, 3), (6, 4), (6, 5), (8, 3)$; the table below states for each cell whether it confirms that record or extends it.

Cell (n, m)	$\text{DIM} = (m-1)^{n+1}$	$\dim_{\mathbb{C}} = \dim_{\mathbb{F}_p}$	Verdict	vs. [1, §5]	Peak RAM
(10, 3)	2,048	1,124	PRIMITIVE	new	14 MB
(8, 4)	19,683	10,730	PRIMITIVE	new	0.72 GB
(8, 5)	262,144	198,640	PRIMITIVE	new	2.57 GB
(6, 5)	16,384	11,484	PRIMITIVE	confirms	0.46 GB
(6, 6)	78,125	59,392	PRIMITIVE	new	0.47 / 0.60 GB
(6, 7)	279,936	235,206	PRIMITIVE	new	2.48 GB
(6, 8)	823,543	720,264	PRIMITIVE	new	2.77 GB
(6, 9)	2,097,152	1,907,032	PRIMITIVE	new	3.71 GB
(4, 4)	243	102	PRIMITIVE	confirms	0.13 GB
(4, 6)	3,125	2,124	PRIMITIVE	confirms	0.13 GB
(4, 10)	59,049	51,288	PRIMITIVE	confirms	0.20 GB
(4, 12)	161,051	145,950	PRIMITIVE	confirms	0.60 GB
(4, 13)	248,832	228,912	PRIMITIVE	new	0.14 GB
(4, 14)	371,293	345,252	PRIMITIVE	new	2.218 GB
(4, 17)	1,048,576	998,016	PRIMITIVE	new	0.79 GB
(4, 19)	1,889,568	1,815,948	PRIMITIVE	new	1.68 GB

Notes. (6, 6) carries two independent modular halves (characteristics 2 and 3), equal partition by partition — to our knowledge the first decided cell whose degree has two distinct prime factors beyond the published table. The prime-degree cells (4, 13), (4, 17), (4, 19) lie beyond the table but inside the region where rational generation is known (Shioda; Aljovin–Movasati–Villaflor); they are census, not candidates. One cell is recorded half-decided for honesty: (4, 15), $\dim_{\mathbb{C}} = 504,924$; the characteristic-5 half closed equal (PRIMITIVE in char 5, peak 3.165 GB); the characteristic-3 half was halted by a no-swap memory discipline before completion — its largest block is the same object measured in (4, 14) char 2, where it closed at exactly $\text{DS}_4(13) = 19,920$.

Since every verdict is PRIMITIVE, each row is a further confirmed instance of Conjecture 1.2 of [1]. Separately, the closed form of [1, Remark 4.4] was verified byte-exact for the $n = 4$ and $n = 6$ rows across a 38-cell sweep of the complex half (degrees well past $m = 12$), with zero outliers.

2 Structure inside the modular half

The expensive half of the criterion is $\dim_{\mathbb{F}_p}$. The following structural facts about that computation were found and proved along the way; full statements, proofs and verifiers are in the repository documents cited.

(a) A boundary of the Jordan-form reduction. The natural fast reduction works in the basis $u = t - 1$, pruning by the relation $u^{m-1} = 0$. That relation is the true ring relation *if and only if m is a prime power*: for composite m with two distinct prime factors, φ factors over $\mathbb{F}_p$, the pruning imposes a false constraint, and the computed rank silently inflates — asymmetrically by characteristic, which is the diagnostic signature. The boundary was found the hard way on (6, 6) and is documented with the failing runs (THE_FRONTIER_6_6); the corrected engine recovers the true verdict in the monomial basis.

(b) The same factorisation as a memory lever. For $p \mid m$, $\varphi = (t - 1)^a g(t)^b$ over $\mathbb{F}_p$ with coprime factors, so the per-variable ring splits by CRT; tensoring over the variables, the module splits into independent blocks the shift respects. Reducing the blocks one at a time and releasing collapses the memory peak from the whole closure to the largest single block (measured factors 0.15–0.39 on (4, 6)). This is what made the composite fourfolds, and the half of (4, 15), fit in 8 GB.

(c) **Torsion localises to one block (proven).** The CRT idempotents are polynomials in the shift, hence central; the module is a direct sum over blocks, its presentation block-diagonal, and the Smith normal form splits per block. Any $\mathbb{Z}$ -torsion of the quotient is therefore confined to a single computable block — the per-block elementary-divisor structure that [3] tabulate around but leave uncomputed.

(d) **The per-block rank is a Degtyarev–Shimada rank at reduced dimension (proven, unconditional).** Writing ℓ for the number of A -pairs a block couples and d for the relevant factor dimension, the rank each block contributes obeys

$$\text{rec}(d, \ell) = \text{DS}_{2(\ell-1)}(d+1) \quad (\ell \geq 2), \quad \text{rec}(d, 1) = d, \quad \text{rec}(d, 0) = 1,$$

verified per block on every dumped cell and confirmed by a blind out-of-sample prediction ((4, 15) char 5, $\dim A = 4$, a value never used in calibration), then proved for every d by a chain of module-dimension identities read from [1] itself: the tensor decomposition in the proof of Corollary 1.7 isolates the core; Remark 4.4 applied one dimension down evaluates it; Lipschitz finiteness carries uniformity in d . Two identifications close the chain, each now a theorem rather than a reading. *First*, the integral block closure is a free $\mathbb{Z}$ -module — the within-pair closure is the ring $\mathbb{Z}[t]/\varphi$, via the cyclotomic identity

$$x \cdot c_0 = -1, \quad c_0 = 1 + x + \cdots + x^{m-2} \quad (\text{since } x^{m-1} = -c_0),$$

so the orbit of the generator regenerates the unit, the closure is $\mathbb{Z}[t]/\varphi$ (free, rank $m-1$), a tensor of free modules is free, and reduction modulo every prime is faithful: the modular computation injects no spurious torsion, in any characteristic. *Second*, the CRT idempotent cut e_A equals the coordinate-zeroing cut of [1, §4.6]: on $V = k[t]/(t^m - 1)$ the generalized eigenspace $E_1(t) = \ker((t-1)^m)$ — the part fixed by the Galois rotation γ_j , i.e. what setting $z_j = 0$ retains under the degree bound $0 \leq \nu < m$ of [1, Lemma 4.1] — equals $\text{im}(e_A)$, in every characteristic, by primary decomposition; the two cuts are the same linear projection, not merely the same count. (In characteristic p the correct reading is the full $(t-1)$ -primary block of dimension p^v , the nilpotent tower; over $\mathbb{C}$, e_A is the Galois averaging projector.)

(e) **Inside one block.** The within-pair coupling is identified functorially: the survivor space is $\text{Sym}^2 V$ on the class layer (proved for odd characteristic; the characteristic-2 obstruction $(\mathbb{Z}/2)^{p^v-1}$ is shown to be the generic torsion of the swap involution, identical for a generic space of the same dimension — hence not a Fermat-specific phenomenon).

3 The exact relation to Conjecture 1.2

Reading [1, §4.5–§4.6] in full: Theorem 1.1(d) presents the coupled module as $\bar{C}_K = (\bigoplus_J \bar{R}_J)/\bar{M}$ with $\bar{M}$ generated by the single element $\sum_J 1_J$; each leg $\bar{R}_J$ is $\mathbb{Z}$ -free by Lemma 4.1; §4.5 establishes the four torsion descriptions of Theorem 1.1 isomorphic; and §4.6 states that the module of the partial Fermat is free *if and only if* Conjecture 1.2 holds in dimension $2s$. With the cut identity of §2(d), that equivalence applies exactly to the blocks measured here: the freedom of the coupling of three or more pairs in recursive dimension $2s$ and Conjecture 1.2 in dimension $2s$ are one statement. Recursive dimensions 0 and 2 sit below the equivalence (free outright — two legs are $\mathbb{Z}[t]/\varphi$ by the identity above, and dimensions 0, 2 are torsion-free by [1] and [2]). Accordingly: this work proves no case of the conjecture beyond those instances the census confirms cell by cell, and claims none; what

it adds is the localisation (any counterexample lives in one identified block, of recursive dimension ≥ 4 , deviating from the value law above), the faithfulness of the modular computation that would measure it, and eleven new confirmed instances.

4 Engineering and reproducibility

All computations ran on a MacBook Air M2 (2022), 8 GB RAM, one thread, throttled to 25% CPU, no swap, exact modular arithmetic throughout (no floating point). The storage cost of an echelon entry was driven from 16 bytes to 1.07 bytes across the engine lineage; the deepest single reduction, (6, 9), held 3.45×10^9 nonzero entries within 3.71 GB. Every number above traces to a raw run log in the repository, every engine is a single self-contained C++ file, and every structural claim has a self-contained Python verifier that recomputes the Degtyarev–Shimada values from scratch:

`github.com/tretoef-estrella`

The repository README states explicitly, for each cell and each claim, what is new, what is confirmation of [1], and what remains open; the five cells inside the published table are labelled as independent reproductions throughout.

References

- [1] A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*, J. Math. Soc. Japan **68**:3 (2016), 975–996. arXiv:1405.4683.
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